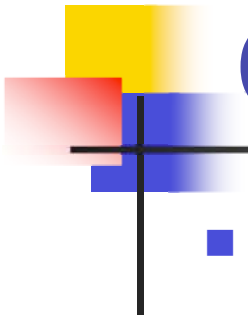


XML at the ASF

Ted Leung

Principal, Sauria Associates, LLC

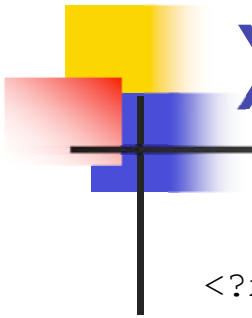
twl@sauria.com



Overview

- xml.apache.org
  - Xerces
  - Xalan
  - FOP
  - Batik
  - Xindice
  - Forrest
  - XML-Security
  - XML-Commons
  - XMLBeans (incubated)
- ws.apache.org
  - XML-RPC
  - Axis
  - WSIF
  - JaxMe (incubated)
- cocoon.apache.org
  - Cocoon
  - Lenya (incubated)

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Xerces-J

```
<?xml version="1.0" encoding="UTF-8"?>
<books xmlns="http://sauria.com/schemas/apache-xml-book/books"
  xmlns:tns="http://sauria.com/schemas/apache-xml-book/books"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation=
    "http://sauria.com/schemas/apache-xml-book/books
     http://www.sauria.com/schemas/apache-xml-book/books.xsd"
  version="1.0">
  <book>
    <title>Effective Java</title>
    <author>Joshua Bloch</author>
    <isbn>yyy-yyyyyyyyyy</isbn>
    <month>August</month>
    <year>2001</year>
    <publisher>Addison-Wesley</publisher>
    <address>New York, New York</address>
  </book>
</books>
```



Xerces-J

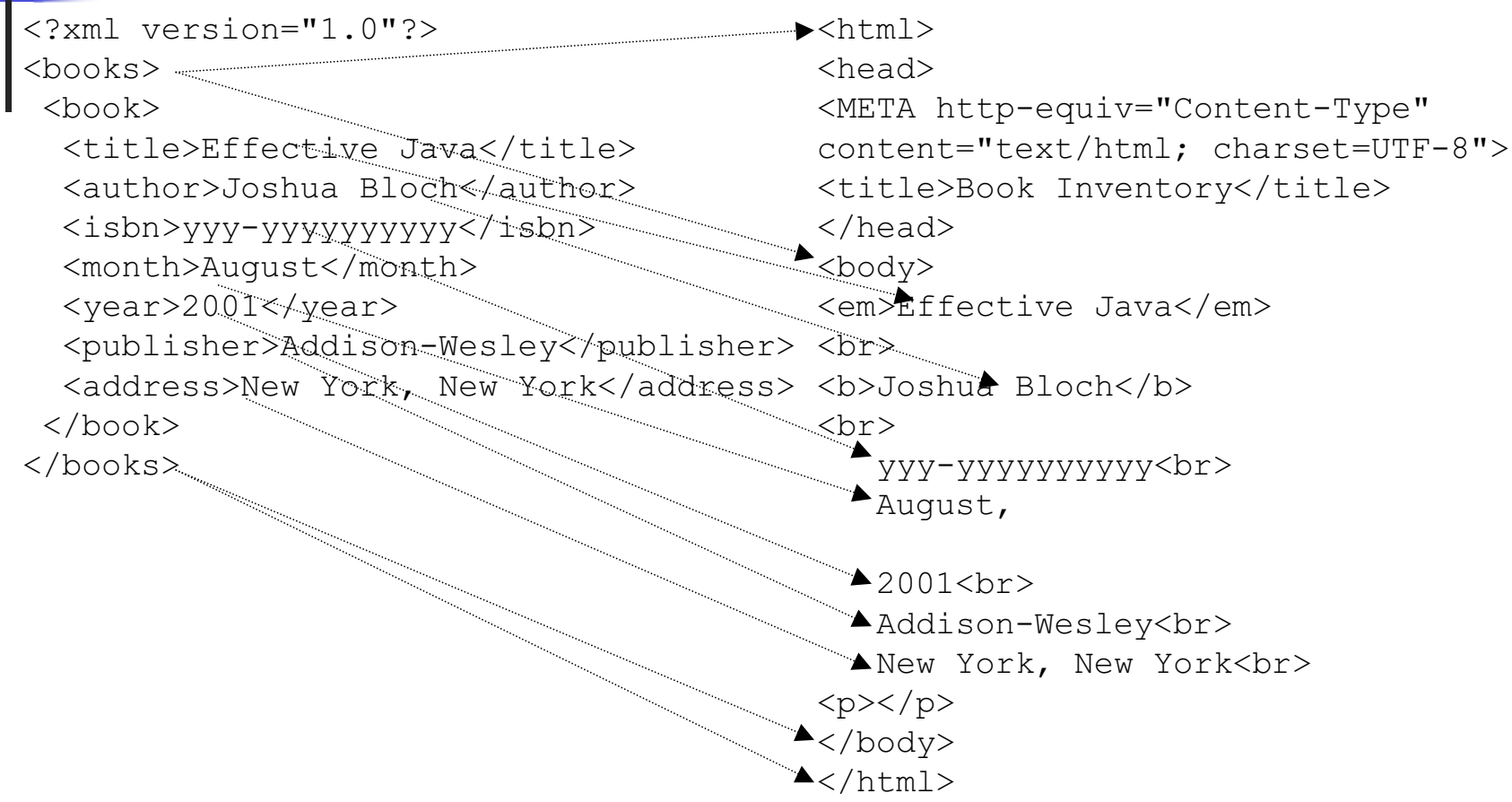
- Provide API's to the parts
  - SAX
  - DOM
  - XNI
  - XNI-based pull
- Validation
  - DTDs
  - XML Schema Support
  - Relax NG support via Andy Clark's Neko Tools for XNI
- When would I use it?
  - Everywhere
- For More
  - M005: Everything you always wanted to know about XML parsing M13:30
  - M019: Apache XML Parser: Xerces2 and Xerces C++ M17:00



Xalan-J

- What does it do?
  - XSLT Processor
- Converts one kind of XML into another
  - Uses a stylesheet (XML document)
  - Stylesheet describes tree transformation
  - Declarative programming model
    - Based on pattern matching

Xalan-J


<?xml version="1.0"?>
<books>
<book>
<title>Effective Java</title>
<author>Joshua Bloch</author>
<isbn>yyy-yyyyyyyyyy</isbn>
<month>August</month>
<year>2001</year>
<publisher>Addison-Wesley</publisher>
<address>New York, New York</address>
</book>
</books>
<html>
<head>
<META http-equiv="Content-Type"
content="text/html; charset=UTF-8">
<title>Book Inventory</title>
</head>
<body>
Effective Java

Joshua Bloch

yyy-yyyyyyyyyy
August,
2001
Addison-Wesley
New York, New York

</body>
</html>

Xalan-J

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0"
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
  xmlns:books="http://sauria.com/schemas/apache-xml-book/books"
  exclude-result-prefixes="books">
    <xsl:output method="html" version="4.0" encoding="UTF-8"
        indent="yes" omit-xml-declaration="yes"/>
    <xsl:template match="books:books">
      <html>
        <head><title>Book Inventory</title></head>
        <body>
          <xsl:apply-templates/>
        </body>
      </html>
    </xsl:template>
    <xsl:template match="books:book">
       <xsl:apply-templates/>
      <p />
    </xsl:template>
    <xsl:template match="books:title">
      <em><xsl:value-of select="."/></em><br />
    </xsl:template>
    …
```



Xalan-J

- TrAX API is part of JAXP
- XSLTC Compiler
- Extensions
  - EXSLT
  - Xalan specific
- When would I use it?
  - Convert XML to HTML
  - Convert XML to XML
    - WML
    - Vocabulary translation
- For More:
  - TU11: Using Xalan to achieve application interoperability T13:30



FOP

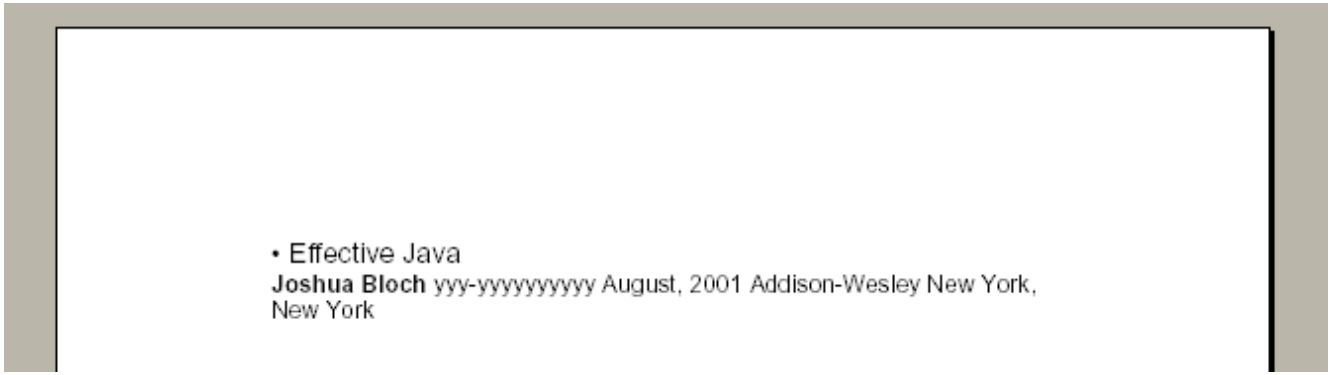
- What does it do?
  - XSL Processor
  - Convert XML with XSL elements into non XML formats

FOP – XSL Input

```
<?xml version="1.0" encoding="UTF-8"?>
<fo:root xmlns:fo="http://www.w3.org/1999/XSL/Format">
 <fo:layout-master-set>
  <fo:simple-page-master margin-right="0.5in" margin-left="0.5in"
      margin-bottom="0.5in" margin-top="0.5in" page-width="8.5in"
      page-height="11in" master-name="book-page">
   <fo:region-body margin="1in"/>
  </fo:simple-page-master>
   fo:layout-master-set>
 <fo:page-sequence master-reference="book-page">
  <fo:flow flow-name="xsl-region-body">
   <fo:block>
    <fo:list-block provisional-label-separation="3pt"
        provisional-distance-between-starts="18pt">
     <fo:list-item>
      <fo:list-item-label>
       <fo:block space-after.optimum="15pt">
         fo:inline font-size="14pt">â€¢ Effective Java</fo:inline>
       </fo:block>
      </fo:list-item-label>
```

FOP – PDF output

```
      <fo:list-item-body>
       <fo:block space-after.optimum="15pt">
        <fo:inline font-weight="bold">Joshua Bloch</fo:inline>
        <fo:inline>yyy-yyyyyyyyyy</fo:inline>
        <fo:inline>August, </fo:inline>
        <fo:inline>2001</fo:inline>
        <fo:inline>Addison-Wesley</fo:inline>
        <fo:inline>New York, New York</fo:inline>
       </fo:block>
      </fo:list-item-body>
     </fo:list-item>
    </fo:list-block>
   </fo:block>
  </fo:flow>
 </fo:page-sequence>
</fo:root>
```





FOP

- Supported formats include
  - PDF
  - PostScript
  - RTF
- Font Handling
- Hyphenation
- When would I use it?
  - Any where you need a format that FOP can produce



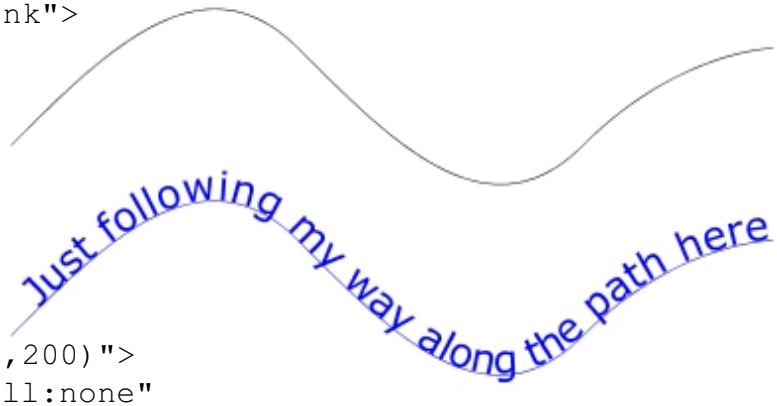
Batik

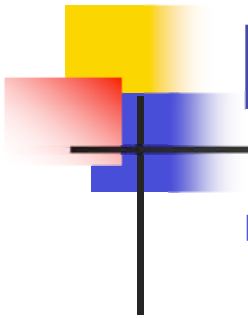
- What does it do?
  - Scalable Vector Graphics (SVG)
    - Vector drawing commands as XML
    - Declarative animation
    - Imperative animation (scripting)

Batik

```
<?xml version="1.0" standalone="no"?>
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 20010904//EN"
  "http://www.w3.org/TR/2001/REC-SVG-20010904/DTD/svg10.dtd">
<svg width="5in" height="3in" viewBox="0 0 800 600"
     xmlns="http://www.w3.org/2000/svg"
     xmlns:xlink="http://www.w3.org/1999/xlink">
 <g id="plainPath">
  <path style="stroke:black; fill:none"
        d="M 0 200
           C 100 100 200   0 300 100
           C 400 200 500 300 600 200
           C 700 100 800 100 800 100"/>
 </g>

 <g id="labelledPath" transform="translate(0,200)">
  <path id="followMe" style="stroke:blue; fill:none"
        d="M 0 200
           C 100 100 200   0 300 100
           C 400 200 500 300 600 200
           C 700 100 800 100 800 100"/>
  <text font-family="Verdana" font-size="42.5" fill="blue" >
   <textPath xlink:href="#followMe" startOffset="40">
    Just following my way along the path here
   </textPath>
  </text>
 </g>
</svg>
```





Batik

- Toolbox
  - SVGGraphics2D
  - JSVGCanvas
  - Browser
  - Image Transcoding
  - Rasterizer
  - Scripting
- When would I use it?
  - Graphical Output – Static or Dynamic
  - Flash-like user interfaces
- For More:
  - M008: Java Applications with Apache Batik
    M 13:30



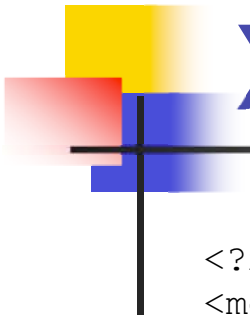
Xindice

- What does it do?
  - Native XML Database
- XML:DB API
  - Collections of XML Documents
  - Searchable by XPath
  - XUpdate
- When would I use it?
  - You want to store XML directly
  - You need XPath query capability



XML-RPC

- What does it do?
  - Use XML to markup RPC's
  - Deliver the XML via HTTP
  - Supports a fixed set of datatypes



XML-RPC

```
<?xml version="1.0" encoding="UTF-8"?>
<methodCall>
 <methodName>createBook</methodName>
 <params>
  <param><value><string>Effective Java</string></value></param>
  <param><value><string>Joshua Bloch</string></value></param>
  <param><value><string>yyy-yyyyyyyyyy</string></value></param>
  <param><value><string>August</string></value></param>
  <param><value><i4>2001</i4></value></param>
  <param><value><string>Addison-Wesley</string></value></param>
  <param><value><string>New York, New York</string></value></param>
 </params>
</methodCall>
```



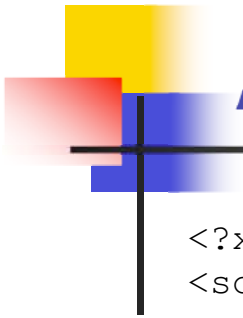
XML-RPC

- When would I use it?
  - You need to talk to another XML-RPC application
- For More:
  - http://www.xmlrpc.com



Axis

- What does it do?
  - SOAP
    - The standardized extensible XML-RPC
  - WSDL
    - A way to describe SOAP services



Axis

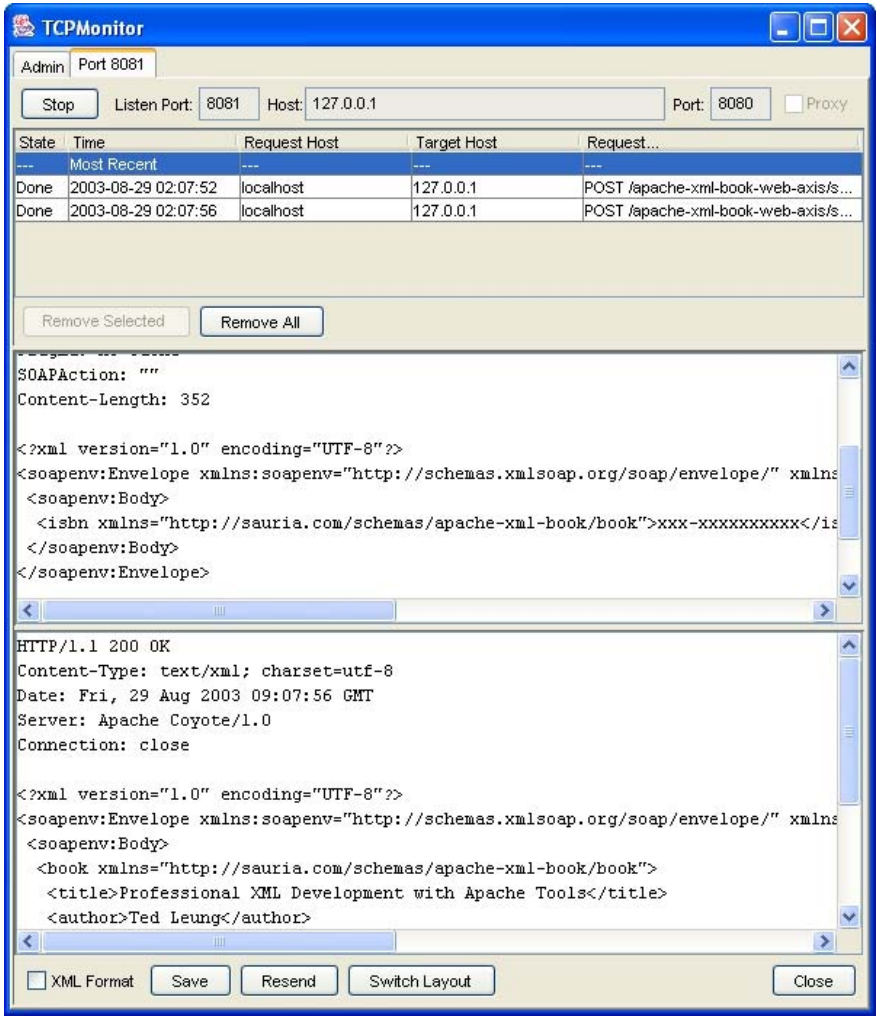
```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope
 xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
 <soapenv:Body>
  <book xmlns="http://sauria.com/schemas/apache-xml-book/book">
   <author>Joshua Bloch</author>
   <title>Effective Java</title>
   <isbn>yyy-yyyyyyyyyy</isbn>
   <month>August</month>
   <year>2001</year>
   <publisher>Addison-Wesley</publisher>
   <address>New York, New York</address>
  </book>
 </soapenv:Body>
</soapenv:Envelope>
```



Axis

- JAX-RPC programming model (JSR-101)
- WSDL tools
  - WSDL2Java
  - Java2WSDL
- TCPMon

Axis


TCPMonitor
Admin
Port 8081
Stop
Listen Port: 8081
Host: 127.0.0.1
Port: 8080
Proxy
State
Time
Request Host
Target Host
Request...
Most Recent
Done 2003-08-29 02:07:52 localhost 127.0.0.1 POST /apache-xml-book-web-axis/s...
Done 2003-08-29 02:07:56 localhost 127.0.0.1 POST /apache-xml-book-web-axis/s...
Remove Selected
Remove All
SOAPAction: ""
Content-Length: 352
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns
<soapenv:Body>
<isbn xmlns="http://sauria.com/schemas/apache-xml-book/book">xxx-xxxxxxxxxx</is
</soapenv:Body>
</soapenv:Envelope>
HTTP/1.1 200 OK
Content-Type: text/xml; charset=utf-8
Date: Fri, 29 Aug 2003 09:07:56 GMT
Server: Apache Coyote/1.0
Connection: close
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns
<soapenv:Body>
<book xmlns="http://sauria.com/schemas/apache-xml-book/book">
<title>Professional XML Development with Apache Tools</title>
<author>Ted Leung</author>
XML Format
Save
Resend
Switch Layout
Close



Axis

- When would I use it?
  - You need a SOAP based web service
  - You need to integrate systems
    - You need to do it over the Internet
  - You need to talk to .NET
- For More:
  - TU15: Building a Web service from SOAP to Nuts T14:30



XML-Security

- What does it do?
  - XML Digital Signature
  - XML Encryption

XML-Security

```
<?xml version="1.0" encoding="UTF-8"?>
<book version="1.0"
 xmlns="http://sauria.com/schemas/apache-xml-book/book"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://sauria.com/schemas/apache-xml-book/book
      http://www.sauria.com/schemas/apache-xml-book/book.xsd">
 <title>Apache XML Tools</title>
 <xenc:EncryptedData Type="http://www.w3.org/2001/04/xmlenc#Element"
  xmlns:xenc="http://www.w3.org/2001/04/xmlenc#">
  <xenc:EncryptionMethod
   Algorithm="http://www.w3.org/2001/04/xmlenc#aes256-cbc"
   xmlns:xenc="http://www.w3.org/2001/04/xmlenc#"/>
  <xenc:CipherData xmlns:xenc="http://www.w3.org/2001/04/xmlenc#">
   <xenc:CipherValue xmlns:xenc="http://www.w3.org/2001/04/xmlenc#">
Pyn0AqcaMaCg5Cv1wy5sDe0I1aox/JzvupzyXcS0AGilXaF4SCbBaiBVS33KxGS8P
   </xenc:CipherValue>
  </xenc:CipherData>
 </xenc:EncryptedData>
 <isbn>xxx-xxxxxxxxxx</isbn>
 <month>October</month>
 <year>2003</year>
 <publisher>John Wiley and Sons</publisher>
 <address>New York, New York</address>
</book>
```



XML-Security

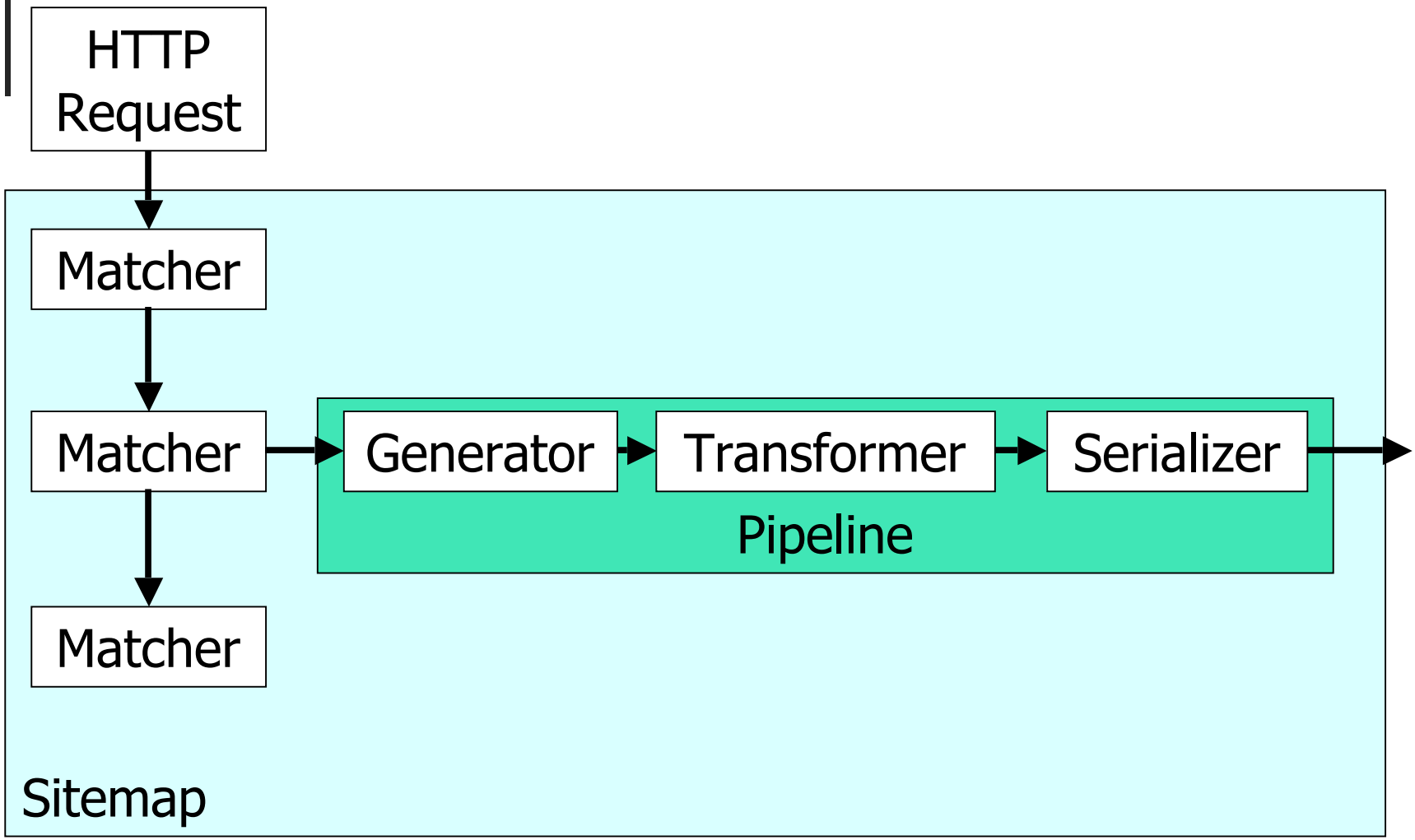
- When would I use it?
  - When you need digital signatures
  - When you need encryption
  - XML based workflows



Cocoon

- What does it do?
  - Web publishing framework
  - Based on XML
  - Not an implementation of a standard
  - Relies on a pipeline architecture

Cocoon



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Cocoon

- When would I use it?
- For More:
  - M013: Introducing Apache Cocoon M16:00
  - M017: Past, present and future of the Apache Cocoon Project M17:00
  - TU16: Porting CMS applications to WebDAV and Cocoon T14:30
  - TU19: The Cocoon Portal T16:00



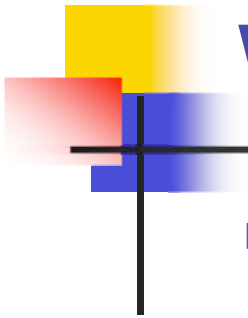
Forrest

- What does it do?
  - Documentation platform based on Cocoon
- Content Templates
  - FAQs, Changelogs, HOWTOs, etc.
- Skins
- Forrestbot
- When would I use it?
  - Used in many xml.apache.org projects



XML Commons

- What does it do?
  - Reusable components
    - SAX and DOM
    - Which
    - Resolver
      - OASIS XML Catalogs
      - OASIS TR9401 Catalogs
      - XCatalog
- When would I use it?



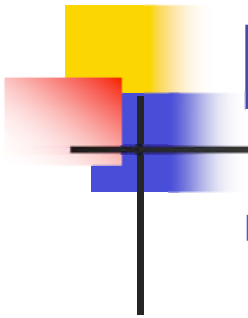
WSIF

- What does it do?
  - Web Services Invocation Framework
  - Allows you to invoke services via a WSDL document
    - Supports
      - Java classes
      - EJB
      - JMS
      - JCA
- When would I use it?
  - If you need a uniform interface to services implemented using different technologies
  - You need isolation from the services API's



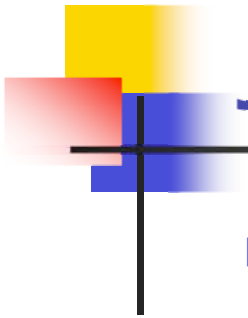
XMLBeans (incubated)

- What does it do?
  - XML instance <-> Java instances
    - Based on XML Schema mapping
    - Populate JavaBeans from XML
    - Generate XML from JavaBeans
  - Cursor based navigation of XML
  - Select XML based using XPath
- When would I use it?
  - You need to map between XML and Java
- For More:
  - M009: Inside Apache XMLBeans M14:30



Lenya (incubated)

- What does it do?
  - Content Management System for Cocoon
    - Revision Control
    - Scheduling
    - Search Engine
    - Staging areas
    - Workflow
    - Browser based WSIWYG Editors
- When would I use it?
  - When you are developing a large site in Cocoon
- For More:
  - TU06: Content Management with Apache Lenya T10:00



JaxMe (incubated)

- What does it do?
  - JAXB implementation
    - Convert XML instances <-> Java Bean instances
  - Store JavaBeans into a database
  - Query that database
- When would I use it?
  - You need to use JAXB



jakarta.apache.org

- Betwixt
  - XML introspection mechanism for mapping beans to XML
- Digester
  - Rule based approach to “digesting” XML configuration files
- JXPath
  - Use XPath to traverse graphs of Java objects



Questions:

- http://xml.apache.org
- http://cocoon.apache.org
- http://ws.apache.org
- http://jakarta.apache.org

- twl@apache.org
- http://www.sauria.com/blog
- XML Development with Apache Tools
  - Wrox Press, December 2003

This presenation has ended.
Click anywhere to return.

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